VIBEVOICE-ASR-STREAMING: WHO SPOKE WHAT IN REAL TIME

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ABSTRACT

Real-time recognition of multi-speaker conversations is becoming increasingly important. Such systems need to transcribe speech, identify the speaker of each segment, and produce speaker-attributed transcripts with low delay. Existing end-to-end speaker-attributed ASR models jointly perform transcription and speaker attribution. However, they generally process complete recordings. Meanwhile, streaming LLM-based ASR has mainly focused on single-speaker transcription and generally lacks speaker-attribution capability. To tackle this issue, we present VIBEVOICE-ASR-STREAMING, an end-to-end framework for streaming speaker-attributed ASR. It interleaves fixed-duration audio chunks with previously generated speaker labels and text in a single autoregressive context, which preserves speaker identities across chunks and enables speaker-attributed transcription in a single process without a separate diarization stage. On five recognition benchmarks, our 7B model achieves the lowest mean recognition error among the compared streaming systems. Across 13 speaker-attributed settings, it achieves the best or tied-best cpWER/cpCER in 12 settings. The expected latency is 2.00 s. We will release the 1.5B and 7B model weights together with the inference code.

1 INTRODUCTION

Recent real-time speech models such as OpenAI's GPT-Live 1 and Google's Gemini 3.8 Live have made low-latency spoken interaction a prominent area of speech technology (OpenAI, 2026; Google, 2026). This growing interest also highlights the importance of recognizing speech accurately, especially in multi-speaker conversations: participants may interrupt one another, alternate rapidly, or return after several minutes, requiring a recognizer to recover not only what was said but also who said it. Streaming speaker-attributed ASR addresses this recognition setting by producing speaker-labeled transcripts while the conversation is still unfolding.

Formally, streaming speaker-attributed ASR must transcribe an ongoing conversation and assign each segment to a consistent speaker identity before the complete recording is available. The task requires speaker identity to remain stable over long intervals: a participant may speak at the beginning of a meeting and return several minutes later, requiring the system to reuse the original label even though the relevant acoustic evidence lies far outside the current chunk.

Three lines of work address parts of this problem. LLM-based recognizers now transcribe long recordings and assign speakers in a single generative pass, including VibeVoice-ASR (Peng et al., 2026b), MOSS Transcribe Diarize (Yu et al., 2026), SoulX-Transcriber (Dai et al., 2026), and SpeakerLM (Yin et al., 2025), but they read the whole recording before emitting output. A second line develops LLM-based ASR for streaming single-speaker transcription. BESTOW (Chen et al., 2024) casts inference as a read–write problem. SpeechLLM-XL (Jia et al., 2024) and Uni-ASR (Xia et al., 2026) consume audio in chunks and carry preceding speech-text context forward. A third line makes multi-talker recognition low-latency, from SURT (Lu et al., 2021; Raj et al., 2023) and t-SOT (Kanda et al., 2022a), which serialize overlapping talkers, to systems that attach speaker identity at low latency through token-level speaker embeddings (Kanda et al., 2022b), an auxiliary speaker branch (Raj et al., 2024), or online diarization cascaded with a recognizer (Han et al., 2021; Liang et al., 2024; Park et al., 2025; Medennikov et al., 2025). Concurrent Speech-LLM systems target the same setting (Shi et al., 2025; Peng et al., 2026a).

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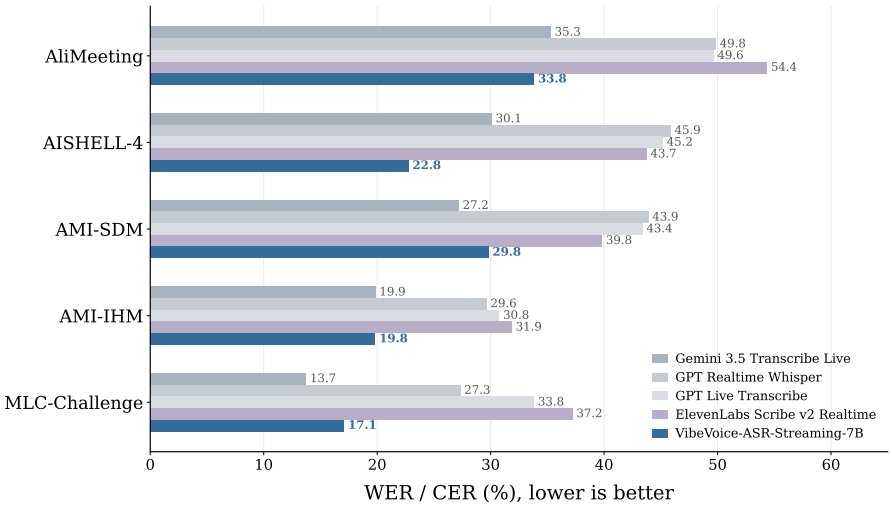


Figure 1: Recognition error (WER/CER; lower is better) on four meeting benchmarks and MLC-Challenge. MLC-Challenge is the macro average over nine languages; Chinese, Japanese, and Korean use CER, and all other datasets and languages use WER.

Despite these advances, existing approaches do not jointly provide low-latency recognition and long-range speaker consistency through a single recognition context. Long-form LLM recognizers wait for complete recordings, while streaming systems either retain context primarily for recognition or rely on separate speaker caches, trackers, or diarization components. As a result, the complete session history does not serve as a unified basis for both transcription and speaker attribution.

We address this limitation with VIBEVOICE-ASR-STREAMING, an end-to-end model that uses the accumulated session history for both recognition and speaker attribution. Following previous streaming Speech-LLMs (Jia et al., 2024; Xia et al., 2026), VIBEVOICE-ASR-STREAMING interleaves incoming audio with generated speaker-attributed text. This design bounds future acoustic context by the chunk contract while retaining the accumulated speech, transcription, and speaker information across chunks without a separate diarization stage. Each chunk is followed by a fixed 4-frame (0.5-s) lookahead. The released 1.5B and 7B models use the 22-frame (2.9-s) chunk configuration, with an expected speaker-attribution latency of 2.00 s. Figure 1 summarizes the recognition-only comparison. As shown in Table 1, the 7B 22-frame configuration achieves the lowest five-setting recognition macro-average among the compared streaming systems. As shown in Table 2, it achieves the best or tied-best speaker-attributed error in 12 of 13 settings across the four meeting benchmarks and nine MLC-Challenge languages.

This paper contributes:

- one of the first end-to-end LLM formulations for streaming speaker-attributed ASR, trained through interleaved speech–text generation to support long-form recognition and speaker attribution. We will release the 1.5B and 7B model weights and inference code.
- a thorough study of streaming speaker attribution, including comprehensive ablations of chunk size, lookahead, model scale, and speaker-label placement, a complete training recipe, and comparisons against the non-streaming model and deployed streaming systems.

2 RELATED WORK

Long-form speaker-attributed ASR with LLMs. SpeakerLM (Yin et al., 2025) is a multimodal LLM that jointly performs speaker diarization and ASR in an end-to-end manner, with a flexible speaker registration mechanism for different registration conditions. MOSS Transcribe Diarize (Yu et al., 2026) formulates speaker-attributed, time-stamped transcription in a unified multimodal LLM with a 128k context window for inputs of up to 90 minutes. VibeVoice-ASR (Peng et al., 2026b) processes up to 60 minutes of audio in a single pass and generates speaker identities, timestamps,

and transcription as a unified rich transcription. SoulX-Transcriber (Dai et al., 2026) jointly models speaker diarization and ASR and uses speaker-aware continuous pre-training followed by supervised fine-tuning to improve speaker discrimination and transcription robustness. These systems process each input offline rather than emitting results incrementally.

Streaming LLM-based ASR. BESTOW (Chen et al., 2024) reformulates streamable Speech-LLM inference as a read–write policy problem within one architecture for offline and streaming speech tasks. SpeechLLM-XL (Jia et al., 2024) is a linear-scaling decoder-only model that processes audio in configurable chunks with a limited attention window and autoregressively generates text for each chunk. Uni-ASR (Xia et al., 2026) unifies non-streaming and streaming LLM-based ASR through joint training, context-aware training, and a fallback decoding strategy.

Streaming multi-talker and speaker-attributed recognition. Streaming multi-talker recognition predates the Speech-LLM era. SURT (Lu et al., 2021) introduced end-to-end streaming multi-talker ASR, with later extensions improving streaming separation, encoding, and speaker attribution across chunks (Raj et al., 2023; 2024). In parallel, t-SOT (Kanda et al., 2022a) serialized tokens from multiple speakers into a single output stream, and its extension incorporated token-level speaker representations for low-latency speaker attribution (Kanda et al., 2022b). Online diarization methods such as BW-EDA-EEND (Han et al., 2021), FS-EEND (Liang et al., 2024), and Streaming Sortformer (Medennikov et al., 2025) progressively introduced incremental processing and explicit mechanisms for online speaker tracking. More recently, JEDIS-LLM (Shi et al., 2025) and G-STAR (Peng et al., 2026a) combined streaming ASR with speaker-aware caches or tracking modules in Speech-LLM-based systems. In contrast, VIBEVOICE-ASR-STREAMING uses no separate speaker cache or tracking module, retaining session-specific speaker identity directly in its interleaved speech–text context.

3 METHOD

3.1 ARCHITECTURE AND STREAMING FORMULATION

Figure 2 illustrates the structure of VIBEVOICE-ASR-STREAMING. Audio is processed by the frozen Acoustic and Semantic encoders of VibeVoice (Peng et al., 2025). The Acoustic encoder follows the σ-VAE design (Sun et al., 2024), while the Semantic encoder extracts content-oriented features on the same temporal grid. Their outputs are concatenated and projected into a Qwen2.5 (Yang et al., 2024) LLM. The encoders downsample 24-kHz audio by $3{,}200\times$, producing 7.5 latent frames per second.

Following streaming Speech-LLMs (Jia et al., 2024; Xia et al., 2026), speech chunks and their generated transcripts are interleaved as $[\widetilde{X}_1, Y_1, \widetilde{X}_2, Y_2, \ldots]$, where $\widetilde{X}_k$ contains the current chunk X_k and its L-frame lookahead. Previous speech and generated text remain in the LLM context, allowing speaker labels to be reused throughout the conversation. For $Y_k = (y_{k,1}, \ldots, y_{k,N_k})$, the generation probability is

$$p(Y_k \mid X_{<k}, \widetilde{X}_k, Y_{<k}) = \prod_{j=1}^{N_k} p\Big(y_{k,j} \mid X_{<k}, \widetilde{X}_k, Y_{<k}, y_{k,<j}\Big) .$$

We use a fixed four-frame (0.5-s) lookahead and evaluate 15-frame (2.0-s) and 22-frame (2.9-s) chunks.

After receiving $\widetilde{X}_k$, the model generates Y_k until emitting <|text_chunk_end|>, which is supervised at every chunk boundary, including those with no target text. Decoding then resumes after the next speech chunk is appended. Optional context, such as names, technical terms, abbreviations, and hotwords, can be prepended and retained throughout the session.

Output format. Speakers are assigned ordinal labels by order of first appearance, and previously introduced labels are reused in later chunks. Concatenating all Y_k yields the final speaker-attributed transcript. Overlapping speech is serialized into consecutive speaker-labeled segments. Appendix A provides the exact syntax and a decoding example. The uncompressed context incurs growing KV-cache memory and per-chunk computation, and the released checkpoints support sessions of up to eight minutes.

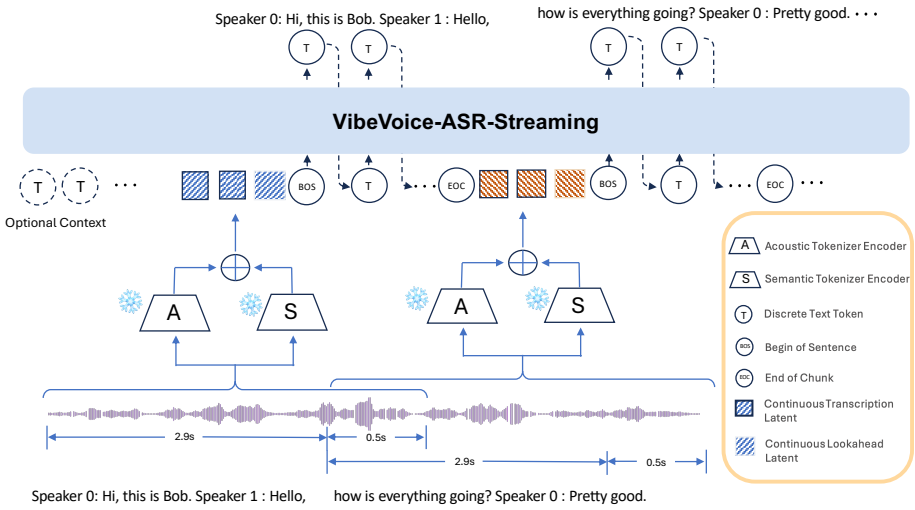


Figure 2: Architecture of VIBEVOICE-ASR-STREAMING. Speech chunks X_k and speaker-attributed text chunks Y_k are interleaved in a single autoregressive context, and each chunk is followed by a fixed $L = 4$-frame (0.5 s) lookahead before its text is generated.

3.2 TRAINING DATA

All training recordings, real and synthetic alike, are prepared the same way: word-level timing is obtained by running Qwen3-ForcedAligner-0.6B (Qwen Team, 2026) over the recording, and the reference transcript is then split into per-chunk targets according to the rule stated in Appendix A.

To improve robustness to multi-speaker conditions and specialized vocabulary, we synthesize meeting-style conversations containing domain terms and proper nouns. The spoken form drives synthesis while the written form remains the ASR target, allowing natural speech but canonical transcription. We then add overlap and convolve the mixture with room impulse responses, updating speaker labels and alignment with the waveform. This yields 50,884 recordings totaling 4,519.6 hours of augmented multi-speaker training speech.

3.3 TRAINING ROUTE

We train VIBEVOICE-ASR-STREAMING in three stages that differ in how training samples are constructed rather than in the model or the training objective.

Stage 1: non-streaming training. The model is first trained with complete recordings to establish speaker-attributed recognition without a streaming constraint.

Stage 2: streaming pre-training. Starting from the Stage-1 checkpoint, we switch to the interleaved form of Section 3.1: each chunk is paired with its speaker-attributed transcription and fixed lookahead. The architecture, trainable modules, and autoregressive objective remain unchanged, so the model adapts to bounded future context without relearning speaker-attributed transcription.

Stage 3: streaming fine-tuning. Stage 2 uses roughly 420,000 hours of speech from the Stage-1 corpus to adapt the model to the streaming format. Stage 3 then fine-tunes on public training splits and

Table 1: Recognition-only streaming ASR results on the four meeting benchmarks and MLC-Challenge. For all online APIs, audio is streamed using 2.9-s chunks, matching the chunk size of VIBEVOICE-ASR-STREAMING. The compared systems are Gemini 3.5 Transcribe Live, GPT Realtime Whisper, GPT Live Transcribe, ElevenLabs Scribe v2 Realtime, and our 7B model. Best results are shown in **bold** and second-best underlined. Chinese, Japanese, and Korean are evaluated with CER, and the remaining languages with WER. The final row reports the mean over the four meeting benchmarks and the MLC-Challenge macro average.

		Gemini 3.5 Transcribe Live[1]	GPT Realtime Whisper [2]	GPT Live Transcribe[3]	Scribe v2 Realtime[4]	Ours 7B
Benchmark	Lang.	Err.	Err.	Err.	Err.	Err.
MLC-Challenge	English	**8.40**	20.66	21.88	22.75	8.44
	French	19.48	27.90	37.83	32.98	**16.42**
	German	**16.96**	33.25	37.35	48.68	21.83
	Italian	**12.98**	28.41	35.72	35.98	17.00
	Japanese	**16.54**	36.50	33.48	33.17	27.85
	Korean	9.59	24.98	32.18	31.53	**9.09**
	Portuguese	**20.83**	38.86	53.46	59.20	28.15
	Russian	**9.66**	20.55	29.38	46.78	15.67
	Spanish	**9.13**	14.88	23.13	23.79	9.31
	Average	**13.73**	27.33	33.82	37.21	17.09
AliMeeting	Chinese	35.29	49.84	49.64	54.35	**33.83**
AISHELL-4	Chinese	30.09	45.86	45.17	43.74	**22.76**
AMI-SDM	English	**27.18**	43.90	43.38	39.81	29.81
AMI-IHM	English	19.85	29.64	30.75	31.86	**19.83**
Average	Mix	25.23	39.31	40.55	41.39	**24.66**

the synthetic multi-speaker data of Section 3.2 to calibrate transcription conventions, speaker-label consistency, and hotword following. Optimizer settings and run scale are given in Appendix A.

Each streaming configuration is initialized from the non-streaming checkpoint of the same scale, avoiding full training from scratch. The chunk size is fixed throughout Stages 2 and 3, and the 15- and 22-frame configurations are trained independently.

4 RESULTS

All models use the frozen Acoustic and Semantic tokenizer encoders and the trainable Qwen2.5 LLM backbone of Section 3.1, and differ only in backbone scale, 1.5B and 7B. Each scale is trained at two chunk sizes, 22 latent frames (2.9 s of audio) and 15 latent frames (2.0 s), under the same fixed 4-frame (0.5 s) lookahead; unless otherwise stated, the reported results use the 7B model with 22-frame chunks.

Datasets. We evaluate on the Chinese meeting corpora AISHELL-4 (Fu et al., 2021) and AliMeeting (Yu et al., 2022), the English AMI-IHM and AMI-SDM conditions (Carletta et al., 2005), and nine aligner-supported languages of MLC-Challenge (Mu et al., 2025). Recordings are split into

[1] https://ai.google.dev/gemini-api/docs/models/gemini-3.5-transcribe

[2] https://developers.openai.com/api/docs/models/gpt-realtime-whisper

[3] https://developers.openai.com/api/docs/models/gpt-live-transcribe

[4] https://elevenlabs.io/docs/api-reference/speech-to-text/v-1-speech-to-text-realtime

[5] https://learn.microsoft.com/en-us/azure/ai-services/speech-service/get-started-stt-diarization

[6] https://docs.cloud.google.com/speech-to-text/docs/multiple-voices

Table 2: Overall streaming speaker-attributed ASR results for the 7B model with 22-frame (2.9 s) chunks. Chinese, Japanese, and Korean are evaluated with CER and cpCER, and the remaining languages with WER and cpWER. For Google STT, D_0 uses the speaker label at first emission, while D_{fin} uses the final revised label after the full recording has been processed. Best results are shown in **bold** and second-best underlined, ranked over the systems that report the metric. A dash marks a language or corpus the system does not support.

		Azure Conversation Transcriber [5]		Google Cloud Speech-to-Text[6]			Ours 7B	
Dataset	Lang.	Err.	cpErr.	Err.	cpErr.$_0$	cpErr.$_f$	Err.	cpErr.
Latency (s)		8.21		–	9.12	51.06	2.00	
MLC-Chal.	English	9.31	23.02	10.57	60.57	28.40	**8.44**	**11.99**
	French	20.55	33.58	–	–	–	**16.42**	**21.50**
	German	**21.12**	27.17	–	–	–	21.83	**24.68**
	Italian	**13.67**	21.53	–	–	–	17.00	**21.20**
	Japanese	**15.04**	**33.01**	–	–	–	27.85	**33.01**
	Korean	10.32	23.97	–	–	–	**9.09**	**23.22**
	Portuguese	**22.76**	**36.11**	–	–	–	28.15	38.25
	Russian	**14.99**	28.11	–	–	–	15.67	**19.47**
	Spanish	11.12	17.00	–	–	–	**9.31**	**11.46**
	AVERAGE	**15.43**	27.06	–	–	–	17.09	**22.75**
AliMeeting	Chinese	**29.40**	52.25	–	–	–	33.83	**39.80**
AISHELL-4	Chinese	24.03	32.51	–	–	–	**22.76**	**28.70**
AMI-SDM	English	33.27	41.40	32.11	73.63	46.65	**29.81**	**39.01**
AMI-IHM	English	24.01	31.90	23.07	71.48	40.15	**19.83**	**27.48**

consecutive non-overlapping segments of at most 480 seconds; no audio is discarded. Single-speaker evaluation uses AISHELL-1 (Bu et al., 2017), LibriSpeech test-clean and test-other (Panayotov et al., 2015), and GigaSpeech (Chen et al., 2021). Only official training splits are included in training, with no evaluation-set overlap.

Metrics. We follow the MeetEval (von Neumann et al., 2023)[7] protocol and report word error rate (WER), which ignores speaker attribution and so reflects recognition quality alone, and concatenated minimum-permutation WER (cpWER), which concatenates the hypotheses and references belonging to each speaker and takes the minimum error over speaker permutations. Chinese, Japanese, and Korean are scored at the character level for every system alike, as CER and cpCER, and columns headed WER and cpWER carry those values on any row or language so scored, including inside the MLC-Challenge average. Speaker-attribution latency is the delay from a spoken word to its settled speaker-attributed transcription. Following each system's output semantics, we report an *expected algorithmic* delay $C/2 + T_{\text{lookahead}}$ for VIBEVOICE-ASR-STREAMING (2.00 s at 22 frames and 1.53 s at 15) and a *measured wall-clock* mean for the cloud services. Appendix B gives the scoring and measurement details.

Compared systems. For recognition-only comparison, Figure 1 and Table 1 additionally include Gemini 3.5 Transcribe Live[1], GPT Realtime Whisper[2], GPT Live Transcribe[3], and ElevenLabs Scribe v2 Realtime[4]. For streaming speaker-attributed recognition, we compare against Microsoft Azure ConversationTranscriber (Azure CT) and Google Cloud Speech-to-Text (Google STT). Google STT revises speaker labels retroactively, so we report two operating points: D_0, using each speaker label when it is first emitted, and D_{fin}, using the final label after the entire recording has been processed. For short-form single-speaker recognition, we compare against open-source streaming ASR models, including Nemotron-3.5, Voxtral-Mini-4B-RT, and X-ASR.

[7] https://github.com/fgnt/meeteval

Table 3: Single-speaker recognition, reported as a class check rather than as a competitive claim. CER for AISHELL-1, WER for the others; the final column is the uniform mean over the four sets. **Bold** marks the lowest value in each column and underline the second lowest.

System	A-1	LS-clean	LS-other	Giga	Avg.
VIBEVOICE-ASR-STREAMING-7B (15 frm)	4.81	2.68	7.40	10.60	6.37
VIBEVOICE-ASR-STREAMING-7B (22 frm)	4.01	2.33	6.49	10.20	**5.76**
Nemotron-3.5 0.6B (NVIDIA, 2026)	13.56	3.02	6.80	11.76	8.79
Voxtral-Mini-4B-RT (Mistral AI, 2026)	9.56	**1.88**	**4.10**	10.11	6.41
X-ASR (GilgameshWind, 2026)	**3.61**	2.96	6.85	**9.64**	5.77

Results. We evaluate VIBEVOICE-ASR-STREAMING from three complementary perspectives: long-form meeting transcription against state-of-the-art streaming ASR APIs, long-form speaker-attributed recognition against leading deployed multi-speaker services, and short-form recognition against open-source streaming ASR models.

Figure 1 and Table 1 test recognition accuracy on long, multilingual meeting recordings, where a streaming system must preserve context while processing audio incrementally. The figure summarizes the five evaluation settings, and the table provides the full per-language comparison with state-of-the-art streaming ASR APIs. VIBEVOICE-ASR-STREAMING is best on AISHELL-4, AliMeeting, and AMI-IHM, while Gemini 3.5 Transcribe Live is best on AMI-SDM. Across the four meeting conditions and the MLC-Challenge macro-average, VIBEVOICE-ASR-STREAMING obtains the lowest average error at 24.66, compared with 25.23 for Gemini 3.5 Transcribe Live, 39.31 for GPT Realtime Whisper, 40.55 for GPT Live Transcribe, and 41.39 for ElevenLabs Scribe v2 Realtime.

Table 2 tests the more demanding setting of streaming speaker-attributed recognition, which requires the system to recognize speech, maintain speaker identities over long recordings, and emit results with low delay. Because no open-source end-to-end speaker-attributed ASR model supports this streaming setting, we compare with leading deployed multi-speaker services and report WER/CER, cpWER/cpCER, and speaker-attribution latency. VIBEVOICE-ASR-STREAMING gives the best or tied-best cpWER/cpCER in 12 of the 13 settings, including eight of the nine MLC-Challenge languages, and improves over Azure CT by 2.39 to 12.45 points on the four meeting benchmarks. Under the reporting definitions in Appendix B, VIBEVOICE-ASR-STREAMING has an expected latency of 2.00 s, compared with measured latencies of 8.21 s for Azure CT and 9.12 s for Google STT; Google STT continues to revise speaker labels for tens of seconds.

Table 3 checks whether the model remains competitive on conventional short-form, single-speaker ASR, a setting outside its primary design target. We compare against open-source streaming ASR models under a common normalization. Although VIBEVOICE-ASR-STREAMING does not rank first on any individual test set, the 22-frame configuration ranks second on AISHELL-1 and both LibriSpeech splits and obtains the best four-set mean by a margin of 0.01. It also outperforms the 15-frame configuration on all four test sets.

VIBEVOICE-ASR-STREAMING performs strongly on both long-form meeting transcription and streaming speaker-attributed recognition. Short-form single-speaker ASR is not its main strength, yet the open-source model remains competitive and achieves the best average result across the four short-form test sets.

5 EXPERIMENTS

Cost of the streaming conversion. Table 4 scores VIBEVOICE-ASR-STREAMING-7B with 22-frame chunks against the non-streaming VibeVoice-ASR checkpoint it is initialized from, scored identically. WER/CER rises by 0.75 to 3.53 points and cpWER/cpCER by 5.13 to 6.67 points on every benchmark. Because cpWER/cpCER reflects both recognition and speaker assignment, its larger degradation is consistent with an additional loss in speaker attribution.

Chunk size and model scale. Under an identical 4-frame lookahead, enlarging the chunk from 15 to 22 latent frames improves both metrics on every benchmark and at both scales (Table 5): on the

Table 4: Cost of the streaming conversion: VIBEVOICE-ASR-STREAMING-7B with 22-frame chunks against the non-streaming VibeVoice-ASR checkpoint it is initialized from, scored identically. Δ is streaming minus non-streaming, so positive values are degradations.

		Non-streaming		VIBEVOICE-ASR-STREAMING		Δ	
Dataset	Lang.	WER	cpWER	WER	cpWER	WER	cpWER
AISHELL-4	Chinese	19.87	22.03	22.76	28.70	+2.89	+6.67
AliMeeting	Chinese	32.15	34.12	33.83	39.80	+1.68	+5.68
AMI-IHM	English	19.08	20.98	19.83	27.48	+0.75	+6.50
AMI-SDM	English	26.28	33.88	29.81	39.01	+3.53	+5.13
MLC-Challenge	Mix	14.08	17.62	17.09	22.75	+3.01	+5.13

Table 5: Effect of chunk size and model scale under an identical 4-frame lookahead, giving expected latencies of 2.00 s at 22 frames and 1.53 s at 15. **Bold** marks the better chunk size for each model; MLC denotes MLC-Challenge.

		7B				1.5B			
		22 frm		15 frm		22 frm		15 frm	
Dataset	Lang.	WER	cpWER	WER	cpWER	WER	cpWER	WER	cpWER
AISHELL-4	Chinese	**22.76**	**28.70**	24.62	32.64	**28.48**	**39.56**	30.33	42.42
AliMeeting	Chinese	**33.83**	**39.80**	34.75	41.66	**39.18**	**48.87**	39.34	51.69
AMI-IHM	English	**19.83**	**27.48**	20.82	35.40	**22.85**	**45.15**	23.39	46.48
AMI-SDM	English	**29.81**	**39.01**	31.88	42.49	**34.43**	**56.83**	35.90	59.49
MLC	Mix	**17.09**	**22.75**	18.52	25.85	**21.81**	**31.15**	24.33	36.01
Average		**24.66**	**31.55**	26.12	35.61	**29.35**	**44.31**	30.66	47.22

five-setting macro-average it is worth 1.46 WER/CER and 4.06 cpWER/cpCER at 7B, and 1.31 and 2.91 at 1.5B. Scale acts the same way: at a fixed chunk size, moving from 1.5B to 7B lowers the mean cpWER/cpCER by 12.76 points at 22 frames and 11.61 at 15, against 4.69 and 4.54 of WER/CER. Both factors therefore have a stronger effect on speaker-attributed error than on recognition-only error. This pattern is consistent with longer chunks and larger backbones providing additional evidence for speaker assignment, although the two effects are not isolated by these experiments. Since 15 frames reduces the expected latency from 2.00 s to 1.53 s, the two chunk settings offer different trade-offs between latency and accuracy.

Speaker-label placement. VIBEVOICE-ASR-STREAMING emits the speaker label before the text of the segment it belongs to, which makes the label available from the first token of a segment. Table 6 compares head and tail label placement under the same 7B, 22-frame, 4-frame-lookahead configuration. They achieve nearly identical mean cpWER/cpCER (31.55 vs. 31.56), while tail placement increases mean WER/CER by 0.98, indicating no benefit from delaying speaker labels. This contrasts with Google STT, whose final label revisions reduce cpWER by 27.0–32.2 points relative to its first-emitted labels (Table 2). One possible explanation is that joint text-speaker generation exploits lexical and conversational context in addition to acoustic evidence, enabling reliable attribution within each chunk.

Lookahead depth. Lookahead (sometimes called right context) supplies no context that a later chunk would not eventually provide; it only lets the model read across a chunk boundary before committing to the text of that chunk. Table 7 varies L over 0, 2, and 4 frames at a fixed 22-frame chunk with everything else held constant, the $L = 4$ column being the released system; both metrics improve strictly with L on all five benchmarks. The two steps cost the same 0.267 s of expected latency. Averaged over the benchmarks, the first pair of frames reduces WER/CER by 1.27 points and cpWER/cpCER by 1.80, while the second pair reduces them by 1.24 and 2.54. The cpWER/cpCER

Table 6: Speaker-label placement: emitting the label before the segment text (head, the released configuration) against emitting it after (tail). Both runs are 7B with 22-frame chunks, a 4-frame lookahead, and the same data; Δ is tail minus head.

		Head (released)		Tail		Δ	
Dataset	Language	WER	cpWER	WER	cpWER	WER	cpWER
AISHELL-4	Chinese	22.76	28.70	24.93	28.96	+2.17	+0.26
AliMeeting	Chinese	33.83	39.80	37.72	40.78	+3.89	+0.98
AMI-IHM	English	19.83	27.48	20.13	28.22	+0.30	+0.74
AMI-SDM	English	29.81	39.01	30.65	36.52	+0.84	−2.49
MLC-Challenge	Mix	17.09	22.75	14.79	23.32	−2.30	+0.57
Average		24.66	31.55	25.64	31.56	+0.98	+0.01

Table 7: Lookahead ablation on the 7B model at a fixed 22-frame chunk. L is the number of future latent frames, and expected latency is $C/2 + L \times 133.3$ ms; each two-frame increase adds 0.267 s.

		$L = 0$ (1.47 s)		$L = 2$ (1.73 s)		$L = 4$ (2.00 s)	
Dataset	Language	WER	cpWER	WER	cpWER	WER	cpWER
AISHELL-4	Chinese	26.07	31.98	24.11	29.16	22.76	28.70
AliMeeting	Chinese	37.02	45.02	35.06	41.07	33.83	39.80
AMI-IHM	English	21.83	34.75	20.97	34.34	19.83	27.48
AMI-SDM	English	33.39	44.05	32.09	43.01	29.81	39.01
MLC-Challenge	Mix	17.60	23.61	17.31	22.85	17.09	22.75
Average		27.18	35.88	25.91	34.09	24.66	31.55
Δ vs. previous column		–	–	−1.27	−1.80	−1.24	−2.54

gain therefore remains substantial at $L = 4$, motivating the four-frame lookahead in the released configuration.

6 Conclusion and Limitations

We present VibeVoice-ASR-Streaming, an end-to-end LLM-based system for streaming speaker-attributed transcription. At 2.00 s expected speaker-attribution latency, the 7B model obtains the lowest five-setting recognition macro-average among the compared streaming systems and the best or tied-best cpWER/cpCER in 12 of 13 settings. Both model scales, vLLM inference code, and an online demo are publicly available.

The current system has three limitations:

- *Multilingual coverage:* The present models cover ten languages because target construction depends on Qwen3-ForcedAligner-0.6B (Qwen Team, 2026). This remains a substantial gap relative to general-purpose ASR systems, so broader deployment requires expanding both forced-alignment and training-data coverage.
- *Long-duration overlap:* Sustained overlapping speech is the largest remaining challenge for streaming speaker-attributed ASR. The decoder must recognize simultaneous speech, preserve each speaker's identity, and serialize the segments into one output stream, causing errors to accumulate during long overlaps. Future work should target this setting directly.
- *First-packet latency:* First-packet latency is a general challenge for LLM-based streaming ASR because the model must accumulate acoustic context before autoregressively generating its initial response. The first output requires 3.5 s with 22-frame chunks and 2.5 s with 15-frame chunks, despite a 2.00 s steady-state expected latency. Reducing this delay without sacrificing recognition or speaker attribution remains an important optimization target.

AI USE STATEMENT

In this work, we used generative AI tools to generate synthetic meeting-style, multi-speaker speech for training. We did not use generative AI tools for other applicable tasks requiring disclosure, including method or experiment design, method implementation, translation, dataset cleaning, or interpretation of results; tasks involving mathematical proofs or qualitative and thematic analysis are not applicable to this work. Additionally, we used generative AI tools to edit the manuscript for clarity and consistency and to assist with formatting and compliance review, including section organization, table and figure captions, references, and disclosure statements. We have reviewed all AI-assisted work, including the synthetic data generation procedure and manuscript edits, for consistency with the reported methods and results. We take responsibility for the final content of this work, including text, claims, or artifacts produced with the aid of generative AI.

ETHICS STATEMENT

This work studies speaker-attributed speech recognition using public research datasets and synthetically generated training speech. We did not collect new human-subject data. Speech and speaker-attribution systems may nevertheless raise privacy, consent, demographic-bias, and surveillance concerns when deployed. Users of the released models should comply with the licenses and consent conditions of their data, evaluate performance across relevant speaker and language groups, and avoid uses that identify or monitor people without appropriate authorization. The authors take responsibility for the research and its stated limitations.

REPRODUCIBILITY STATEMENT

Sections 3 and 4 describe the model, training route, evaluation datasets, metrics, and experimental setup. Appendices A and B document the output construction, optimization settings, scoring protocol, latency definition, and baseline configurations. The released checkpoints and inference code support reproduction of the reported inference and evaluation procedures.

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A Output Format, Target Construction, and Training Details

The speaker label is written in plain text as `\n Speaker k:`; no timestamps are emitted, and a chunk containing no speech yields an empty Y_k. A word belongs to chunk k if its end time falls at or before the boundary of chunk k, within a 0.1 s tolerance.

The listing below shows an example output for the released 22-frame configuration, with one chunk arriving every 2.9 s.

```
chunk 0 : " \n Speaker 0:Let's start with the deployment plan."
chunk 1 : " \n Speaker 1:First, we should verify the latency."
chunk 2 : " \n Speaker 0:Agreed. We can run that today."
chunk 3 : " \n Speaker 1:Should we check multilingual results?"
chunk 4 : ""
chunk 5 : " \n Speaker 0:Yes, especially French and German."
chunk 6 : " \n Speaker 1:Okay. I'll prepare the scripts."
chunk 7 : "We can compare the results tomorrow."
chunk 8 : " \n Speaker 0:Please save the per-file outputs."
chunk 9 : " \n Speaker 1:Will do. \n Speaker 0:Great."
chunk 10: "Let's review them when the runs finish."
```

Stages 2 and 3 use AdamW (β_1=0.9, β_2=0.95, weight decay 0.1), gradient clipping at 2.0, bfloat16, a cosine schedule with peak learning rate 5×10^{-5}, and sequences packed to 8,192 tokens. Stage 2 is a multi-node run over the 420,000-hour corpus; Stage 3 is a 500-step run on eight GPUs at a global batch of 64 sequences with 35 warmup steps, and the released checkpoints are taken at step 400.

B Evaluation Protocol and Baseline Systems

This appendix is organized into three parts: the common evaluation and runtime protocol, the recognition-only baselines used in Tables 1 and 3, and the speaker-attributed baselines used in Table 2.

B.1 Evaluation and Runtime Protocol

Scoring. Chinese, Japanese, and Korean are scored at the character level using CER and cpCER; every other language is scored at the word level using WER and cpWER. All systems are scored with the same normalization and the same MeetEval cpWER implementation, with every utterance counted.

Latency. VibeVoice-ASR-Streaming is reported as the expected algorithmic delay $C/2 + T_{\text{lookahead}}$. For the cloud services we report a measured wall-clock delay:

$$\bar{\ell} = \frac{\sum_j n_j \, (\ell_j + d_j/2)}{\sum_j n_j}, \tag{1}$$

where ℓ_j is the delay between the end of the audio span covered by emitted result j and the time its speaker label settles, n_j and d_j are its word count and duration, and $d_j/2$ is the expected position of a word inside it. Word counts follow the scoring tokenization, and all baseline audio is pushed at no faster than $1\times$ real time.

Real-time factor. The expected algorithmic latency assumes that a chunk is decoded before the next one is complete, so we verify this assumption for the more demanding 7B 15-frame configuration. Decoding a chunk takes 146–208 ms against a 2,000 ms chunk, giving an RTF of at most 0.104 (Table 8).

B.2 Recognition-Only Streaming Baselines

Online API versions in Table 1. We evaluate Gemini 3.5 Transcribe Live, GPT Realtime Whisper, GPT Live Transcribe, and ElevenLabs Scribe v2 Realtime, using the service versions named in the

Table 8: Measured serving cost of the 7B 15-frame configuration on one A100 80GB PCIe under vLLM in bfloat16, batch size one; median of three runs on AMI-IHM audio, the two longest inputs obtained by tiling a 237 s recording. Per-chunk compute stays well below the 2000 ms chunk at every length.

Audio (s)	Chunks	Decode (s)	RTF	ms / chunk
30	15	2.90	0.097	193
60	30	4.83	0.081	161
120	60	8.78	0.073	146
237	119	20.13	0.085	169
360	180	32.54	0.090	181
480	240	49.94	0.104	208

table and linked in its column headings. Each API receives audio in 2.9-s increments, matching the 22-frame configuration of VIBEVOICE-ASR-STREAMING, without text prompts, hotword lists, or contextual strings. We otherwise retain the provider's default decoding settings.

Public streaming models in Table 3. All rows are our own measurements under a single normalization. X-ASR uses its released `chunk-1920ms` streaming configuration. Voxtral-Mini-4B-Realtime uses `transcription_delay_ms=2400` (2.4 s), its longest configurable delay. Nemotron-3.5-ASR uses `att_context_size=[56,13]` (1.12 s). These are the released settings closest to our chunk sizes. Nemotron additionally receives the `zh-CN` language identifier on AISHELL-1 and `en-US` elsewhere; our models receive no language identifier.

B.3 SPEAKER-ATTRIBUTED STREAMING BASELINES

Table 9 summarizes the side information supplied to the systems in Table 2.

Table 9: Side information supplied to each system at inference time. VIBEVOICE-ASR-STREAMING receives none of it.

System	Language prior	Speaker count	Hotwords / context
Azure CT	✓ explicit locale	–	–
Google STT	✓ `en-US` only	✓ oracle count	–
VIBEVOICE-ASR-STREAMING	–	–	–

Azure CT. We run `ConversationTranscriber` through version 1.51.1 of the Python SDK `azure-cognitiveservices-speech` against an `S0`-tier resource in `eastus2`. Each file runs in its own session from 16 kHz mono 16-bit PCM pushed through a `PushAudioInputStream`, and we consume only the final `transcribed` events. The language prior is `zh-CN` for AISHELL-4 and AliMeeting, `en-US` for AMI, and the per-utterance locale for MLC-Challenge, with `pt-PT` for Portuguese; no speaker count is supplied. The 8.21 s of Table 2 follows Equation 1.

Google STT. We call Google Cloud Speech-to-Text v1 (`google.cloud.speech_v1`) through the `StreamingRecognize` RPC, using LINEAR16 16 kHz mono audio, the `en-US` language code, and no explicit `model` value. Interim results, automatic punctuation, word time offsets, and speaker diarization are enabled, with `min_speaker_count` set to 2 and `max_speaker_count` to the reference speaker count of each recording. Coverage is English only: AMI-IHM, AMI-SDM, and the English portion of MLC-Challenge, 30.80 h and 256,419 reference words in total. D_0 and D_{fin} carry identical text and differ only in speaker attribution, and the latency reported for Google STT times the settling of speaker labels rather than of text. Text settles after about 1.5 s, a word's speaker label keeps changing for a further 16.5–31.3 s (per-recording medians), 42.6–67.9% of words have their label revised at least once after first emission, and within a 30 s revision budget 66.4% (AMI-IHM), 66.8% (AMI-SDM), and 78.4% (MLC English) of labels agree with D_{fin}.